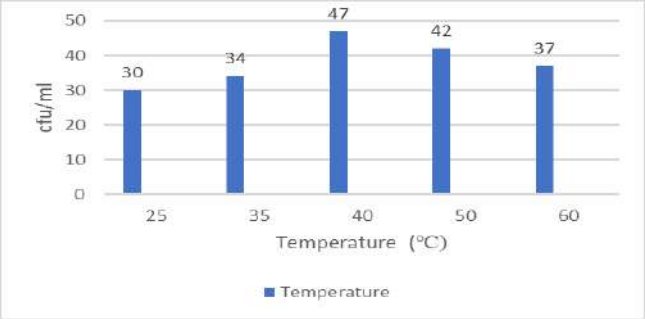


Fig. 3- Effect of temperature on bacterial growth

DISCUSSION

Enzymological and growth studies have revealed the enzymes producing ability of many bacteria. These enzymes are responsible for digestion and providing immunity to its host. Recent studies have also put a light on the capability of bacteria for utilizing Reactive Oxygen Species (ROS), like hydrogen peroxide (H_2O_2), and for its signalling and survival.[13] Reactive Oxygen Species (ROS) are a group of molecules that are produced in the cell through metabolism of oxygen. Endogenous Reactive Oxygen Species (ROS) such as hydrogen peroxide (H_2O_2) have been recognized as a destructive molecule.[14]

Watt *et al.* (2004)[15] have documented that hydrogen peroxide (H_2O_2) causes toxic effect in many ways, some of which are lipid peroxidation and corrosive damage. However, in 1970s, a small collection of studies reported that exogenously added hydrogen peroxide (H_2O_2) could mimic the signalling activity of some hormones.[16] Gough and Cotter (2011)[14] have also reported that hydrogen peroxide (H_2O_2) can modulate enzyme activities by altering its working mechanism which is known as oxidative modulation.

Catalase is an enzyme that is also found in biological systems. It actually catalyses the hydrogen peroxide (H_2O_2) decomposition which ultimately converts it into non-toxic compounds i.e. water and oxygen. It is a well known enzyme for protecting cell from oxidative damage.[17]

$$2H_2O_2 \xrightarrow{\text{Catalase enzyme}} 2H_2O + \underset{\text{Air bubbles}}{O_2\uparrow\uparrow}$$

Whitten bury (1964)[18], stated that bacteria like *Bacillus*, *Leuconostoc*, *Streptococci* and *Pediococci* have the capability of producing catalase enzyme. This shows resemblance with the result obtained in present investigation in which the isolated bacteria was found to be a *Bacillus* species which is also catalase positive.

So far, the reaction velocity is concerned; it has extremely high turnover number which means one molecule of catalase can convert millions of hydrogen peroxide (H_2O_2) molecule into water and oxygen per second.[19,20] This shows resemblance with the present experiment in which brisk effervescence was observed which is the indication of high reaction rate of hydrogen peroxide (H_2O_2) and catalase.

Catalase enzyme has a great role in protecting the cells and other biological components from damage. However, Zamocky *et al.* (2008)[21] opined that only selective organisms have catalase producing ability. In some studies, it has also been reported that catalase activity is acid resistant, which means catalase activity will not get affected by stomach acid if the bacteria is fed as probiotic.[18,22]

On the other hand, Kirchman *et al.* (2005)[23] opined that temperature plays a significant role in the growth and metabolism of bacteria. Membré *et al.* (2005)[24] added that effect of temperature can vary from species to species of microbes. On examining the effect of temperature on the isolated bacteria, it was observed that 40°C was the optimum temperature for its growth. So far, the commercial aspect is concerned, catalase enzyme has a significant role in textile industries as an agent for removing excess hydrogen peroxide from fabric. Catalase is also enormously used for food preservation. It is also in use for elimination of oxygen from wine before bottling.[25,26]

Considering the multiple benefits of this catalase enzyme and due to the easy handling of bacteria, the isolated bacteria can be used for mass production of enzyme catalase in cost efficient manner.

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Blood biochemical profile in carbohydrate metabolism in desi rabbit of Madhepura district of Bihar

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Abstract- The desi rabbit of Madhepura district was selected for biochemical study of the blood parameters. It was found that the total protein content of honey treated rabbits was found to be increased significantly as compared to control, suggesting anabolic effect of honey in rabbit. However no effect on sex was observed. There is highly significant effect of honey treatment is event as increased level of blood glucose was found which provide energy and non-significant effect on sex was observed. There is significant effect of honey on total cholesterol content among group's increases but non-significant effect on sex was observed. In total lipid content show non-significant effects on group but have significant effect on sex. There is non-significant effect of honey on serum phospholipids among group and sex.

Key words: Desi Rabbit, Biochemical blood parameters, carbohydrate metabolism

INTRODUCTION

India is a vast country. The rapid growth of the human population results in continued widening up the gap between demand and supply of the required quantity and quality of food. The problem of malnutrition resulting from food shortage is bound to grow more and more acute with the increase in the human population, to bridge the wide gap between large requirements and low animal protein availability. It is essential to improve and multiply all meat-producing animals in the country.

The rearing of domestic rabbits is an established micro livestock industry in many countries. In many European and Asiatic countries, rabbits are domesticated for meat. Broiler breeds of rabbits have also been introduced in India to explore its avenue as an alternative source of animal protein. The physiological responses of animals are almost crucial in a better understanding of environmental stress and adaptability, which will significantly enhance managerial capability.

Rabbit farming is a widespread livelihood practice across the country. This livestock industry is becoming popular in Bihar also due to its ease of business. Rabbits are best known for being prolific; these herbivores efficiently convert fodder to food. The motive behind meat production is to convert plant proteins of little use to people as food into high-value animal protein. Inefficient production systems, rabbits can turn 20 per cent of the proteins they eat into edible meat. Comparable figures for other species are 22 to 23 per cent for broiler chickens, 16 to 18 per cent for pigs and 8 to 12 per cent for beef.

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Rabbits are reared for their meat, but their hair and skin are also important. Their meat is rich in protein and can be part of a good diet. They have very little fat and are almost cholesterol-free. They are palatable, easily digestible, and consumed in any weather condition, making them a good source of nutrition.

The rabbit meat becomes more prevalent among people requiring less fat, cholesterol, and sodium. So it is not harmful to the heart patient. The cholesterol content of Rabbit meat is lower than other meat; it is suggested because of the inter-relationship between cholesterol, saturated and polyunsaturated fatty acid and caloric value.

The magnitude and trend of rabbit mortality in various seasons and age groups are vital for commercial rabbit farming. Proper nutrition is one of the essential aspects of broiler rabbit production. The protein and energy content of the diet plays a vital role in rabbit nutrition.

The information on biochemical blood profiles for cholesterol, lipid and protein in desi rabbits of Bihar is lacking. The present studies were made to see the effect on these parameters supplemental with antistress or immunomodulatory agent like honey on Rabbit, which will give an additional guide line for future planning or general health status of the Rabbit. Honey is a natural sweetish substance prepared by honey-bee from the nectar of flower contains majority (70-75%) of simple sugars as glucose and fructose, water (18-20%) small amount of sucrose (3-4%) and trace (1-2%) of vitamins, mineral, enzymes, acids and unknown factors including antibacterial effects. In addition, it serves as an instantaneous source of energy, vitamins and trace mineral.

Biochemical characterization of Rabbits in India or tribal belt will help better understand Rabbit in relation to growth, meat, fur and wool quality. In perspective, the present study was undertaken with 'active to assess the gross biochemical constituents including the enzymes of serum in control and honey treatment group and to characterize genetically in desi rabbits available in Madhepura through biochemical polymorphism study'.

MATERIALS & METHODS

Desi rabbits six each of either sex, were divided into two groups. One is honey treated group and the other is the control group. Group II had given 5% simple sugar level in drinking water. All the Rabbit were kept under uniform management practices till the end of the experiment. Blood samples were collected at the end of the investigation directly from Rabbit's heart with or without anticoagulants into a sterile test tube. Serum was harvested from whole blood and kept refrigerated until analyzed. Biochemical parameters were studied with the help of separated serum. Estimation of total blood glucose: The Nelson Somogyi method (1945) estimated blood glucose. 0.1 ml of blood in a test tube was taken, and 9.5 ml of zinc sulphate solution was added and mixed by rotation. Then alkaline copper reagent was mixed by tapping the top of the tube. It was covered with a marble. It was placed upright in a boiling water bath for 5 minutes. Then 1 ml arsenomolybdate coloring reagent was added and mixed, and then it was diluted to 10 ml with distilled water. A blank and standard was run simultaneously. Then absorbance was recorded at 540 nm in photoelectric colorimeter model AE-11 Japan.

Estimation of serum cholesterol: Total serum cholesterol in blood was estimated by method of Zlatkis, *et.al*. (1953)[1]. 0.1 ml serum sample was taken in a test tube. Glacial acetic acid 6.0 ml and coloring reagent 0.4 ml were added. The contents were mixed well. Simultaneously, 0.1 ml of water as blank and 0.1 ml of standard cholesterol solution 2 mg /ml were run similarly treated. Then tubes were allowed to cool at room temperature. The optical density was recorded at 540 nm in photoelectric colorimeter model AE-11 Japan.

Estimation of serum lipid: The total serum lipid estimation was done by using the method of Mandal (2002)[2]. 1 ml of serum was taken with 19 ml of chloroform:methanol (2:1 v/v) mixed well and allowed to stand overnight. The extract was filtered by using Watsman no.1, filter paper in 50 ml of conical flask. Filter paper with residue was cut into pieces and again transferred into 19 ml chloroform kept overnight. This procedure was repeated 3 times. The filtrate was combined with previous extract.

The extract was evaporated at 45°C - 50°C in Sand bath for breaking the volume of original lipid extract in chloroform:methanol:water (64:32:4 v/v) and evaporated to dryness in Sand bath at 45°C - 50°C. This step was repeated thrice and dried residue was dissolved in chloroform:methanol (2:1 v/v) and transferred to separating funnel for removing nonlipid impurities. The